

Computational Algebraic Geometry

Lecture III - Finding Gröbner Basis

Juliette Bruce (Dartmouth)

• Def: Fix a monomial order $\leq$ on $k[x_1, \dots, x_n]$. If $I \subseteq k[x_1, \dots, x_n]$ is an ideal then the initial ideal of I with respect to $\leq$ is

$$\text{in}_{\leq}(I) = \langle \text{LT}_{\leq}(f) \mid f \in I \rangle.$$

A Gröbner basis for I with respect to $\leq$ is a finite set $(g_1, \dots, g_s) \in I$ with

$$\langle \text{LT}_{\leq}(g_1), \dots, \text{LT}_{\leq}(g_s) \rangle = \text{in}_{\leq}(I).$$

Three Questions: ① Is every ideal in $k[x_1, \dots, x_n]$ finitely generated?

(i.e. $I = \langle g_1, \dots, g_s \rangle$ for some polynomials g_i)

(2) $\Rightarrow$ (1)

② Does every ideal have a Gröbner basis?

③ If a Gröbner basis exist how can we find one?

• Thm: (Hilbert's Basis Theorem): Every ideal in $k[x_1, \dots, x_n]$ is finitely generated.

(Hilbert's Basis Theorem, E. Noether): If R is Noetherian ring then

$R[x]$ is Noetherian



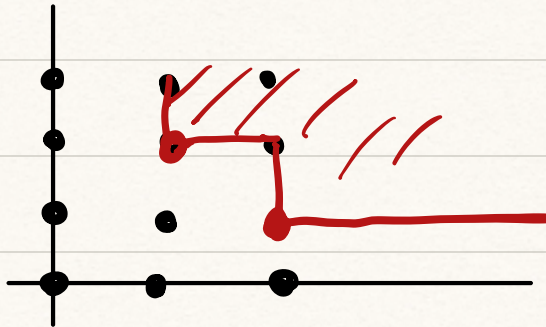
every ideal is f.g. or ACC on ideals

• Notice every initial ideal $\text{in}_\leq(I)$ is a monomial ideal

i.e. generated by monomials

ideal membership
is entirely division

• Lemma: (Dickson's Lemma): If $A \subseteq \mathbb{Z}_{\geq 0}^n$ then A has finitely many elements that are minimal w.r.t. the componentwise partial order.



• Given $A \subseteq \mathbb{Z}_{\geq 0}^n$ we can form a monomial ideal

$$I_A = \langle x^a \mid a \in A \rangle \subseteq k[x_1, \dots, x_n]$$

Cor: Every monomial ideal is finitely generated, I_A is generated by the minimal elements of A .

Thm: If $\leq$ is a monomial order and $I \subseteq k[x_1, \dots, x_n]$ then

I has a gröbner basis w.r.t. $\leq$.

Pf: ① $\text{in}_\leq(I)$ is a monomial ideal

② $\langle m_1, \dots, m_t \rangle = \text{in}_\leq(I)$ w/ m_i monomial (Dickson's Lemma)

* ② There exists $g_i \in I$ w/ $LT(\theta_i) = m_i$ (*)

③ conclude $\theta_1, \dots, \theta_t$ is a GB.

• Def: (S-Polynomials): The S-polynomial of two non-zero polynomials f and g is

$$S(f, g) = \frac{\text{lcm}(LT(f), LT(g))}{LT(f)} f - \frac{\text{lcm}(LT(f), LT(g))}{LT(g)} g$$

• Thm: (Buchberger's Criterion) : Suppose $\langle \theta_1, \dots, \theta_s \rangle = I$
 $G =$

Then $(\theta_1, \dots, \theta_s)$ is a Gröbner basis for I if and only if

$$\text{rem}_G(S(\theta_i, \theta_j)) = 0$$

↑ gives an algorithm for computing GB's.

• Ex: (Affine Twisted Cubic) $\mathbb{K} \longrightarrow \mathbb{K}^3 \quad t \longmapsto (t, t^2, t^3)$

$$\langle x^2 - y, xy - z \rangle = \mathbb{K}[x, y, z]$$

GB w/ $\prec$ GRevLex

$$x^2 - y, \quad xy - z, \quad y^2 - xz$$

$$\begin{pmatrix} 1 & x & y \\ x & y & z \end{pmatrix} \nearrow 2 \times 2 \text{ minors}$$

• Def: Projective n -space over a field K is the set of all lines through the origin in K^{n+1} denoted $\mathbb{P}^n$

$$\begin{aligned}\mathbb{P}^n &= \{ \text{lines thru the origin of } K^{n+1} \} \\ &= \{ 1\text{-dim subspaces of } K^{n+1} \} \\ &= (K^{n+1} \setminus \{0\}) / \sim\end{aligned}$$

where $\sim$ is the equivalence relation $v \sim w \iff v = \lambda w$ for $\lambda \in K^\times$.

$$\mathbb{P}^n = \{ [a_0 : a_1 : \dots : a_n] \}$$

where $[a_0 : a_1 : \dots : a_n] = \text{equivalence class of } (a_0, \dots, a_n)$
 $= \{ \lambda (a_0, \dots, a_n) \mid \lambda \in K^\times \}$

Ex: $\mathbb{P}^2$ $[1:0:0]$ and $[2:0:0]$ and $[4:0:0]$ are the same.

• Lemma: There is a bijection

$$K^n \xrightarrow{\iota_0} U_0 \subseteq \mathbb{P}^n$$

$$(a_1, \dots, a_n) \longmapsto [1, a_1, a_2, \dots, a_n]$$

$$U_0 = \{ [1: a_1: \dots: a_n] \}$$

- Def: A polynomial $f \in K[x_0, \dots, x_n]$ is homogenous of degree d iff every monomial in f has degree d . An ideal is homogenous iff it can be generated by homogenous polynomials.

- Lemma: Let f be homogenous of degree d and $\lambda \in K^n$

$$f(\lambda x_0, \dots, \lambda x_n) = \lambda^d f(x_0, \dots, x_n)$$

- Def: If $J \subseteq S = K[x_0, \dots, x_n]$ is a homogenous ideal then define the projective variety associated to J

$$V_{\mathbb{P}^n}(J) = \{ [a_0: \dots: a_n] \in \mathbb{P}^n \mid f(a_0, \dots, a_n) = 0 \ \forall f \in J \}$$

- Def: If $X \subseteq K^n$ is a variety the projective closure of X is the smallest projective variety containing $\psi_0(X)$ where $K^n \xrightarrow{\sim} U_0 \subseteq \mathbb{P}^n$.

- Ex: $X \subseteq K^3$

$$\langle x^2 - y, xy - z \rangle = K[x, y, z]$$

GB w/t GRevLex

$$x^2 - y, \quad xy - z, \quad y^2 - xz$$

Projective Closure of $X \subseteq \mathbb{P}^3$ $\mathbb{K}[x, y, z, w]$

$V_{\mathbb{P}^3}(J)$ for some homogenous ideal J

$$J = \langle x^2 - wy, xy - zw, y^2 - xz \rangle$$

↑
~~projective~~ twisted cubic